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|-------------|---------------|------------------|
| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |



**GCSE – NEW**

3445UB0-1



**APPLIED SCIENCE (Double Award)**  
**UNIT 2: Space, Health and Life**

**HIGHER TIER**

**MONDAY, 11 JUNE 2018 – MORNING**

**1 hour 30 minutes**

| For Examiner's use only |             |              |              |
|-------------------------|-------------|--------------|--------------|
|                         | Question    | Maximum Mark | Mark Awarded |
| Section A               | 1.(a)(b)(c) | 19           |              |
|                         | 1.(d)       | 6            |              |
| Section B               | 2.          | 16           |              |
|                         | 3.          | 12           |              |
|                         | 4.          | 12           |              |
|                         | 5.          | 10           |              |
|                         | Total       | 75           |              |

3445UB01  
01

**ADDITIONAL MATERIALS**

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

Question **3(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

**Section A**

*Answer **all** questions in the spaces provided.*

*Use the information in the separate Resource Folder to answer the following questions.*

1. (a) Use the information in **Table 1** to answer the following questions.

- (i) Europa is an asteroid. Estimate its temperature and orbital period. [2]

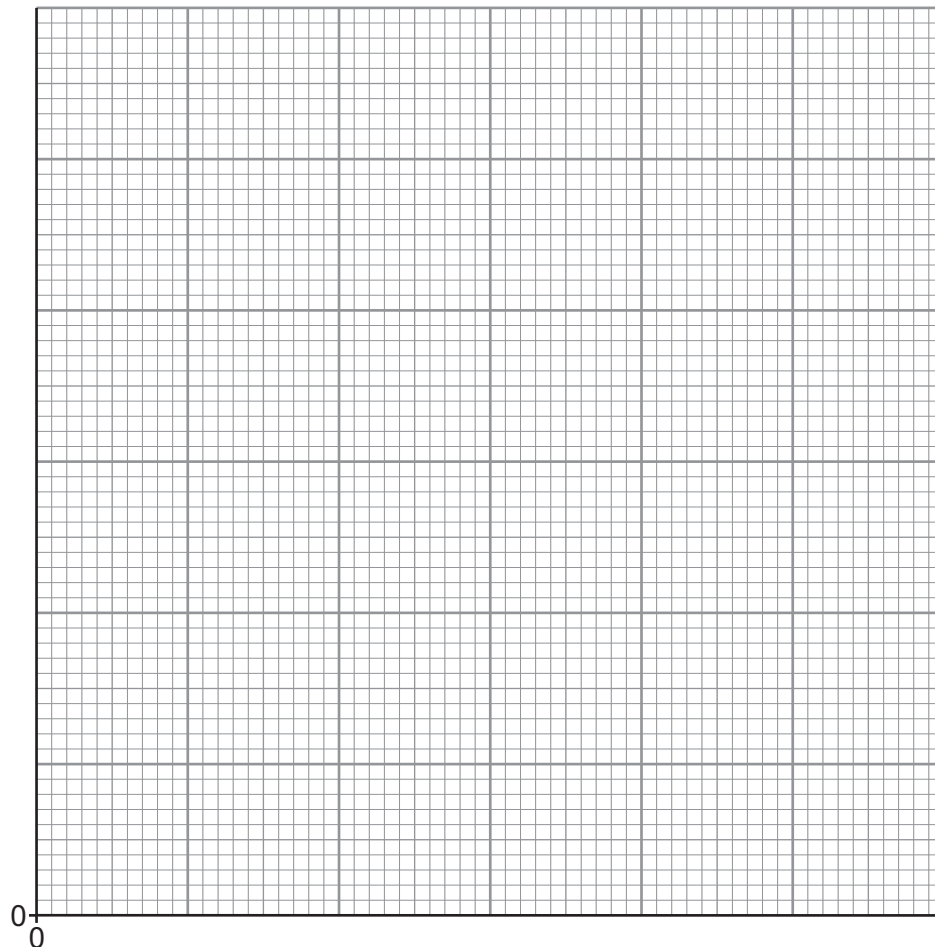
Temperature = ..... °C

Orbital period = ..... days

- (ii) Pluto is no longer classed as a planet. State **one** reason why. [1]

- (iii) Plot the points on the grid below to show how the orbital velocity of the five planets Mercury, Venus, Earth, Mars and Jupiter depends on distance from the Sun. Draw a suitable line. [4]

Orbital  
Velocity  
(km/s)



Distance from Sun (AU)

- (iv) Orbital velocity is not proportional to the distance from the Sun.  
Explain how your graph shows this to be true. [2]

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- (b) Use your knowledge and the information on **pages 4 and 5** to answer the following questions.

- (i) Explain why sunspots appear darker than the surrounding area of the Sun. [2]

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- (ii) State **two** ways in which human activities may be affected by solar flares. [2]

1. ....

2. ....

- (iii) Explain why the effect of solar flares on the Earth is not the same for each sunspot cycle. [2]

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- (c) Use the information in **Diagram 1** to describe the differences between Aristotle's model of the Solar System and the 2006 model. [4]

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(d) Use the information in **Diagram 3** to answer the following questions.

(i) Explain what you can deduce about the motion of the three galaxies. [3]

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(ii) Explain what you can deduce about the chemical composition of the three galaxies. [3]

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**Section B**

2. Velothon Wales is a weekend of road cycling events. The Bloodwise cycling team raises funds to help beat blood cancer.

(a) Three cyclists are monitored before the Velothon. They have blood samples taken.

The results of their blood tests are shown in the table below. Normal ranges are also given.

| Name of cyclist     | Red blood cell count<br>( $10^6/\text{cm}^3$ ) | White blood cell count<br>( $10^3/\text{cm}^3$ ) | Platelet count<br>( $10^3/\text{cm}^3$ ) |
|---------------------|------------------------------------------------|--------------------------------------------------|------------------------------------------|
| <i>Normal range</i> | <i>4.4 - 5.8</i>                               | <i>3.9 - 10.8</i>                                | <i>130 - 400</i>                         |
| Brian               | 5.5                                            | 35.1                                             | 240                                      |
| Gareth              | 2.2                                            | 6.6                                              | 115                                      |
| Jonathan            | 4.7                                            | 8.8                                              | 320                                      |

- (i) Suggest which cyclist needs treatment for an infection. [2]

Cyclist .....

Reason .....

.....

- (ii) Suggest which cyclist needs treatment to raise their haemoglobin level. [2]

Cyclist .....

Reason .....

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- (iii) Suggest which cyclist will experience a problem if he has a fall and cuts himself. [2]

Cyclist .....

Reason .....

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- (b) Explain how the structure of veins and capillaries are related to their function. [4]

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- (c) (i) During a training ride, Jonathan accelerated from rest over a distance of 10 m in 2.5 s then travelled another 96 m at a constant speed in 6 s.  
Use the equations:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration of the cyclist.

[4]

$$\text{Acceleration} = \dots\dots\dots \text{ m/s}^2$$

- (ii) Jonathan has a height of 1.9 m. He is advised his maximum BMI should be 22.  
Use the equation:

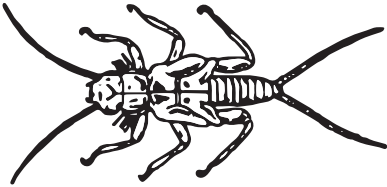
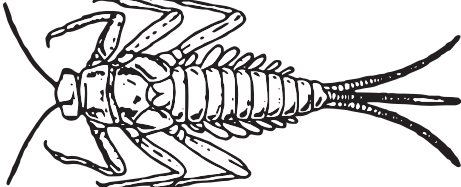
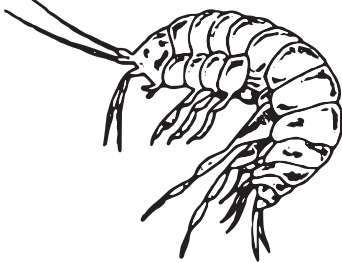
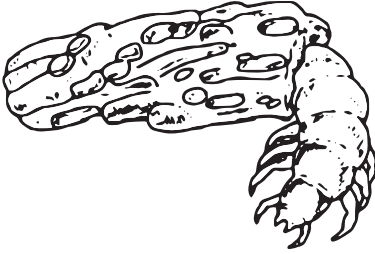
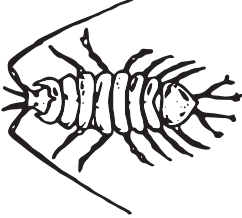
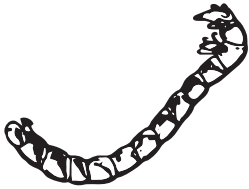
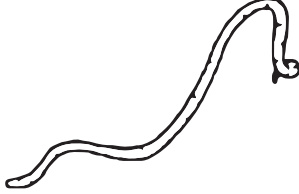
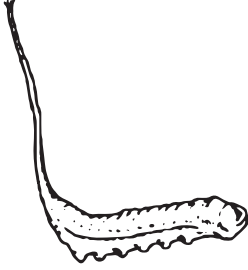
$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

to calculate his maximum mass.

[2]

$$\text{Mass} = \dots\dots\dots \text{ Kg}$$

3. Water pollution can be monitored using invertebrates as indicators. Some indicator species are shown below.

| Water quality       | Indicator species                                                                                                                                                         |                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Clean water         |        | <p>Stonefly nymph</p> <p>Mayfly nymph</p>       |
| Some pollution      |      | <p>Freshwater shrimp</p> <p>Caddisfly larva</p> |
| Moderate pollution  |   | <p>Water louse</p> <p>Bloodworm</p>             |
| High pollution      |   | <p>Sludgeworm</p> <p>Rat-tailed maggot</p>      |
| Very high pollution | No life                                                                                                                                                                   |                                                 |

(Not drawn to scale.)



- (a) Describe how you would carry out an investigation to determine water quality in a stream using invertebrates as indicator species. [6 QER]

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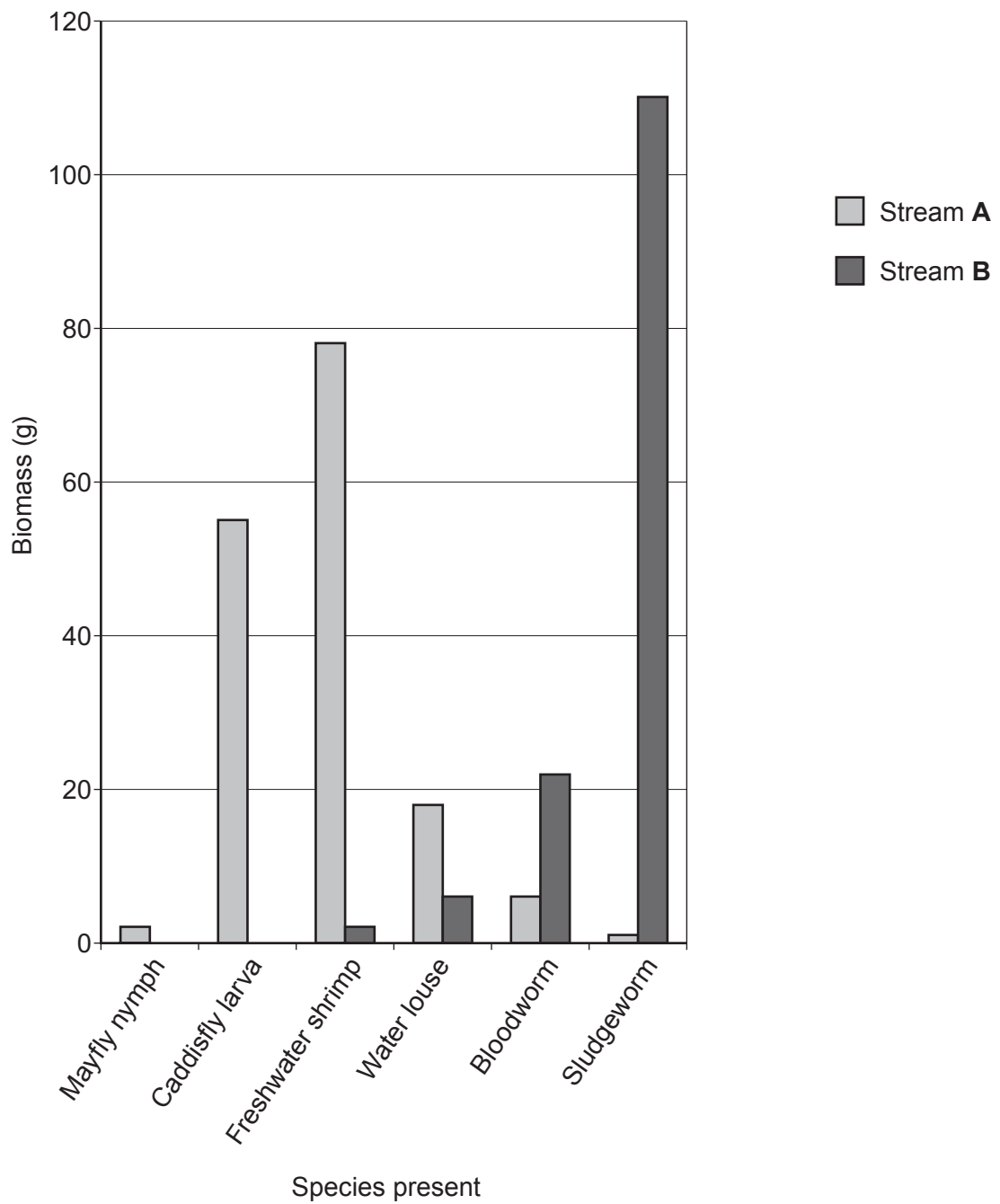
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(b) The results from such an investigation are shown in the chart below.



- (i) Explain what conclusions can be made about the water quality in stream **A** and stream **B**. [4]

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- (ii) Explain why it is difficult to obtain repeatable data in this experiment. [2]

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4. (a) Green plants capture a small percentage of solar energy during the process of photosynthesis. The table below shows how the absorption of light by a green plant depends on the wavelength of the visible light.

| Wavelength of visible light (nm) | Absorption (%) |
|----------------------------------|----------------|
| 400                              | 85             |
| 450                              | 100            |
| 500                              | 2              |
| 550                              | 12             |
| 600                              | 18             |
| 650                              | 35             |
| 700                              | 75             |

- (i) Describe how the absorption of light depends on the wavelength of visible light. [3]

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- (ii) Electromagnetic waves travel at  $3 \times 10^8$  m/s in space.

Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the frequency of light which is most strongly absorbed. [3]

$$(1 \text{ nm} = 1 \times 10^{-9} \text{ m})$$

Frequency = ..... Hz

- (b) The table shows what happens to the energy taken in each day by organisms in the food chain below.

grass → grasshopper → shrew → fox

| Organisms   | Energy per day (MJ) |                             |                 |
|-------------|---------------------|-----------------------------|-----------------|
|             | As waste            | Released during respiration | Used for growth |
| grass       | 6                   | 10                          | 8               |
| grasshopper | 14                  | 22                          | 3               |
| shrew       |                     | 26                          | 4               |
| fox         | 24                  | 32                          | 4               |

- (i) The shrew releases 60% of its energy during respiration and for growth. Calculate how much energy is released by the shrew as waste.

[2]

Energy as waste = ..... MJ

- (ii) Explain why the amount of energy released during respiration by each organism increases through the food chain.

[2]

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- (iii) Sometimes plant seeds will stick to the coat of a fox. Explain how this benefits the plant.

[2]

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5. Nuclear medicine involves the use of radioactive substances in the diagnosis and treatment of disease. Nuclear medicine records radiation emitted from within the body rather than radiation that is generated by external sources like X-rays. Nuclear medicine scans differ from radiology as the emphasis is not on imaging but on organ function.

Information about some isotopes that emit gamma ( $\gamma$ ) rays is given in the table.

| Radioisotope  | Half-life   | Energy of $\gamma$ rays |
|---------------|-------------|-------------------------|
| caesium-137   | 30.17 years | 0.662 MeV               |
| cobalt-60     | 5.26 years  | 1.17 MeV                |
| iodine-125    | 59.6 days   | 31.4 keV                |
| thallium-201  | 73.0 hours  | 71.0 keV                |
| palladium-103 | 17.0 days   | 21.0 keV                |

- (a) (i) Describe the nature of a gamma ( $\gamma$ ) ray.

[2]

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- (ii) Explain why radioisotopes that emit gamma rays are suitable for examining organ function when injected into the body.

[2]

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- (iii) Explain which is the most suitable radioisotope from the table to inject into the human body as part of a gamma camera investigation.

[2]

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- (b) In order to treat prostate cancer, palladium-103 pellets are placed directly into the prostate gland. They remain permanently in place. The gamma emissions from the pellets are almost undetectable when their activity drops to  $\frac{1}{32}$  of its original value. Calculate the time taken for this reduction to occur. [4]

Time = ..... days

**END OF PAPER**

Examiner  
only

10

# THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

|                                                      |  |                                                      |  |                                                                                                                                                                                                                                                          |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |
|------------------------------------------------------|--|------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|
| <div><div>1<br/>H<br/>Hydrogen<br/>1</div></div>     |  |                                                      |  |                                                                                                                                                                                                                                                          |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     | <div><div>4<br/>He<br/>Helium<br/>2</div></div>     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |
| <div><div>7<br/>Li<br/>Lithium<br/>3</div></div>     |  | <div><div>9<br/>Be<br/>Beryllium<br/>4</div></div>   |  | <div><div>11<br/>B<br/>Boron<br/>5</div><div>12<br/>C<br/>Carbon<br/>6</div><div>14<br/>N<br/>Nitrogen<br/>7</div><div>16<br/>O<br/>Oxygen<br/>8</div><div>19<br/>F<br/>Fluorine<br/>9</div><div>20<br/>Ne<br/>Neon<br/>10</div></div>                   |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |
| <div><div>23<br/>Na<br/>Sodium<br/>11</div></div>    |  | <div><div>24<br/>Mg<br/>Magnesium<br/>12</div></div> |  | <div><div>27<br/>Al<br/>Aluminium<br/>13</div><div>28<br/>Si<br/>Silicon<br/>14</div><div>31<br/>P<br/>Phosphorus<br/>15</div><div>32<br/>S<br/>Sulfur<br/>16</div><div>35.5<br/>Cl<br/>Chlorine<br/>17</div><div>40<br/>Ar<br/>Argon<br/>18</div></div> |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |
| <div><div>39<br/>K<br/>Potassium<br/>19</div></div>  |  | <div><div>40<br/>Ca<br/>Calcium<br/>20</div></div>   |  | <div><div>45<br/>Sc<br/>Scandium<br/>21</div></div>                                                                                                                                                                                                      | <div><div>48<br/>Ti<br/>Titanium<br/>22</div></div>  | <div><div>51<br/>V<br/>Vanadium<br/>23</div></div>   | <div><div>52<br/>Cr<br/>Chromium<br/>24</div></div>   | <div><div>55<br/>Mn<br/>Manganese<br/>25</div></div>  | <div><div>56<br/>Fe<br/>Iron<br/>26</div></div>       | <div><div>59<br/>Co<br/>Cobalt<br/>27</div></div>   | <div><div>59<br/>Ni<br/>Nickel<br/>28</div></div>     | <div><div>63.5<br/>Cu<br/>Copper<br/>29</div></div> | <div><div>65<br/>Zn<br/>Zinc<br/>30</div></div>     | <div><div>70<br/>Ga<br/>Gallium<br/>31</div></div>   | <div><div>73<br/>Ge<br/>Germanium<br/>32</div></div> | <div><div>75<br/>As<br/>Arsenic<br/>33</div></div>   | <div><div>79<br/>Se<br/>Selenium<br/>34</div></div>   | <div><div>80<br/>Br<br/>Bromine<br/>35</div></div>   | <div><div>84<br/>Kr<br/>Krypton<br/>36</div></div> |
| <div><div>86<br/>Rb<br/>Rubidium<br/>37</div></div>  |  | <div><div>88<br/>Sr<br/>Strontium<br/>38</div></div> |  | <div><div>89<br/>Y<br/>Yttrium<br/>39</div></div>                                                                                                                                                                                                        | <div><div>91<br/>Zr<br/>Zirconium<br/>40</div></div> | <div><div>93<br/>Nb<br/>Niobium<br/>41</div></div>   | <div><div>96<br/>Mo<br/>Molybdenum<br/>42</div></div> | <div><div>99<br/>Tc<br/>Technetium<br/>43</div></div> | <div><div>101<br/>Ru<br/>Ruthenium<br/>44</div></div> | <div><div>103<br/>Rh<br/>Rhodium<br/>45</div></div> | <div><div>106<br/>Pd<br/>Palladium<br/>46</div></div> | <div><div>108<br/>Ag<br/>Silver<br/>47</div></div>  | <div><div>112<br/>Cd<br/>Cadmium<br/>48</div></div> | <div><div>115<br/>In<br/>Indium<br/>49</div></div>   | <div><div>119<br/>Sn<br/>Tin<br/>50</div></div>      | <div><div>122<br/>Sb<br/>Antimony<br/>51</div></div> | <div><div>128<br/>Te<br/>Tellurium<br/>52</div></div> | <div><div>127<br/>I<br/>Iodine<br/>53</div></div>    | <div><div>131<br/>Xe<br/>Xenon<br/>54</div></div>  |
| <div><div>133<br/>Cs<br/>Caesium<br/>55</div></div>  |  | <div><div>137<br/>Ba<br/>Barium<br/>56</div></div>   |  | <div><div>139<br/>La<br/>Lanthanum<br/>57</div></div>                                                                                                                                                                                                    | <div><div>179<br/>Hf<br/>Hafnium<br/>72</div></div>  | <div><div>181<br/>Ta<br/>Tantalum<br/>73</div></div> | <div><div>184<br/>W<br/>Tungsten<br/>74</div></div>   | <div><div>186<br/>Re<br/>Rhenium<br/>75</div></div>   | <div><div>190<br/>Os<br/>Osmium<br/>76</div></div>    | <div><div>192<br/>Ir<br/>Iridium<br/>77</div></div> | <div><div>195<br/>Pt<br/>Platinum<br/>78</div></div>  | <div><div>197<br/>Au<br/>Gold<br/>79</div></div>    | <div><div>201<br/>Hg<br/>Mercury<br/>80</div></div> | <div><div>204<br/>Tl<br/>Thallium<br/>81</div></div> | <div><div>207<br/>Pb<br/>Lead<br/>82</div></div>     | <div><div>209<br/>Bi<br/>Bismuth<br/>83</div></div>  | <div><div>210<br/>Po<br/>Polonium<br/>84</div></div>  | <div><div>210<br/>At<br/>Astatine<br/>85</div></div> | <div><div>222<br/>Rn<br/>Radon<br/>86</div></div>  |
| <div><div>223<br/>Fr<br/>Francium<br/>87</div></div> |  | <div><div>226<br/>Ra<br/>Radium<br/>88</div></div>   |  | <div><div>227<br/>Ac<br/>Actinium<br/>89</div></div>                                                                                                                                                                                                     |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      |                                                    |
|                                                      |  |                                                      |  |                                                                                                                                                                                                                                                          |                                                      |                                                      |                                                       |                                                       |                                                       |                                                     |                                                       |                                                     |                                                     |                                                      |                                                      |                                                      |                                                       |                                                      | Key                                                |

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|-------------------------|
| 1<br>H<br>Hydrogen<br>1 |
|-------------------------|

Key

|        |                      |
|--------|----------------------|
| $A_r$  | relative atomic mass |
| Symbol |                      |
| Name   |                      |
| $Z$    | atomic number        |